KHUSHI BAGGA

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SUMMARY

Computer Science Engineering student specializing in Artificial Intelligence and Future Technologies (CGPA 9.1/10) with hands-on experience building AI-powered applications, native cross-platform software, and interactive web products. Proficient in Python, Java, JavaScript, Swift, React, and FastAPI, with practical exposure to Large Language Models, AI agents, and prompt engineering. Delivers projects end to end and independently — from protocol-level systems programming to responsive front-end interfaces — and competes in national-level hackathons. Focused on generative AI, human-AI interaction, and responsible AI development.

EDUCATION

Chitkara University — Punjab, India
Bachelor of Engineering (B.E.), Computer Science Engineering — Artificial Intelligence & Future Technologies
2025 – Present | Expected Graduation: 2029 | CGPA: 9.1/10.0 (Semesters I & II)
Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Python Programming, Java Programming, Object-Oriented Programming, Linear Algebra, Calculus, Digital Electronics & Computer Architecture, Foundations of Computer Science

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, Swift, HTML5, CSS3
Frontend Development: React (React 19), SwiftUI, Responsive Web Development, Framer Motion, GSAP, UI/UX Development
Backend & APIs: FastAPI, REST API Design, API Integration, Socket Programming
AI & Machine Learning: Large Language Models (LLMs), LangChain, AI Agents, Prompt Engineering, Vector Search (Pinecone), Responsible AI
Databases & Cloud Services: Supabase, Firebase, Pinecone, LocalStorage
Tools & Platforms: Git, GitHub, Xcode, Figma, Blender, ADB (Android Debug Bridge), macOS, Android
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture, Concurrency, Client-Server Architecture

PROJECTS

PUL — Android-to-Mac Universal Device Integration

Swift, SwiftUI, macOS, Android, VideoToolbox, H.264, AVFoundation, ADB, Socket Programming

- Built two native macOS applications in Swift that turn an Android phone into an extension of the Mac — a mirrored, fully controllable screen and a hardware input device.
- Reimplemented the scrcpy wire protocol client from scratch in Swift, decoding H.264 with VideoToolbox and rendering through AVSampleBufferDisplayLayer for hardware-accelerated, low-latency mirroring.
- Engineered full bidirectional control: cursor tap, drag, swipe, scroll and long-press, Mac keyboard input, two-way clipboard synchronization, and Opus/AAC audio forwarding decoded into AVAudioEngine.
- Delivered wireless Wi-Fi connectivity with automatic reconnection and capped exponential backoff, plus continued mirroring while the phone display is powered off.
- Extended the platform with PUL Device: a multi-touch trackpad with gesture recognition, haptic feedback, media-key injection, ~80 Hz accelerometer and gyroscope streaming, and a live H.264 camera feed.
- Debugged low-level socket synchronization and connection race conditions to establish reliable concurrent video, audio, and control channels.

SAHAAS — Crowdsourced Civic & Mental Health Support Platform

React, FastAPI, Firebase, Supabase, Large Language Models, IVRS

- Architected a multi-channel platform for surfacing early signals of civic and mental health issues and routing users to appropriate interventions.
- Designed an ingestion pipeline supporting both web-based reporting and IVRS voice input to reach users across varying levels of digital access.
- Built an AI-assisted workflow for signal identification, prioritization, and continuity of support across follow-ups.

- Produced the full solution architecture, technical approach, impact model, and feasibility strategy presented for hackathon evaluation.

Student Support & Wellbeing AI Agent

AI Agents, Large Language Models, Prompt Engineering, Responsible AI

- Designed a conversational AI assistant that helps university students manage academic workload, deadlines, and everyday stress.
- Developed contextual prompting strategies that produce empathetic, personalized responses rather than generic model output.
- Evaluated responsible AI considerations including context retention, user privacy, emotional dependency risk, and appropriate escalation boundaries for AI-based support.

CommonGround — Community Hobby Platform

React, JavaScript, UI/UX Design, Figma, User Research

- Designed and prototyped a community platform connecting college students and young adults through shared hobbies and interests.
- Defined a feature set spanning hobby discovery, community groups, mentor interaction, workshops, in-app chat, and location-based exploration.
- Conducted early-stage user research to identify barriers to learning new hobbies and translated findings into product and interface decisions.
- Implemented a responsive, accessible interface with a lightweight design system focused on usability and community engagement.

EXPERIENCE & LEADERSHIP

EvolveAI, Chitkara University — Punjab, India

Student Member, Technical & Creative Contributor | 2025 – Present

- Contribute to AI-focused student initiatives spanning technical projects, product ideation, and visual communication.
- Develop and present applied AI concepts and technology-focused content to peer and faculty audiences.
- Collaborate across technical and creative teams to take project ideas from concept through to presentation.

HACKATHONS & COMPETITIONS

Smart India Hackathon (SIH) — Participant

- Developed a technology-driven solution addressing real-world civic and social challenges.
- Owned problem analysis, solution architecture, technical approach, value proposition, impact analysis, feasibility assessment, and final presentation.

ADDITIONAL INFORMATION

Areas of Interest: Artificial Intelligence, AI Ethics, Generative AI, AI Agents, Human-AI Interaction, Software Development, Emerging Technologies, Data Structures & Algorithms, UI/UX

Languages: English, Hindi